USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 30 LAB - CFAR Processing
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 4.7: I can apply constant false-alarm rate (CFAR) processing to radar data and evaluate detection performance.

This packet is the L30 Lab, in three parts, and you turn it in once, complete. Do **Part 1** before you go to the bench: the predictions you will test. **Part 2** is your own measurements, with the Phaser app's CFAR threshold overlay on your own kit, and three short tasks in the course's notebook, run on range-Doppler bursts. Do **Part 3** after the bench: compare the measurements with the predictions and explain the differences. You work this lab on your own, outside class, following the procedure on the L30 page.

Due: see Gradescope.

Grading. The packet is checked and returned with feedback. It is scaffolding for your final report, not part of the report: the report explains the radar in your own words, and text lifted from this packet earns nothing there.

The radar is the PHASER's FMCW mode with the L29 burst: 256 chirps of $300\ \mu$s, one every 0.5 ms, $B = 500$ MHz at 10 GHz, so a range-Doppler cell is 0.30 m by 0.12 m/s. The detector is cell-averaging CFAR unless a question says otherwise: the threshold is α times the mean power of the N reference cells.

Part 1: Pre-Lab (individual, before the bench)

1. **The bias from the false-alarm probability.**

   (a) The notebook's 2D window has 2 guard cells and 4 reference cells on each side of the cell under test, in range and in Doppler. Count its reference cells, N.

   N = _____

   (b) Fill in the bias, in dB, for each window and design P_{fa}. The first row is the lesson's one-dimensional window, 2 guard and 16 reference cells a side.

Window	N	$P_{fa} = 10^{-3}$	$P_{fa} = 10^{-4}$	$P_{fa} = 10^{-6}$
16 a side, range only	32			
4 each way, 2D				

(c) Show that as N grows without limit the bias for $P_{fa} = 10^{-6}$ approaches the 11.4 dB threshold of Lesson 27, and explain in one sentence why a window with fewer reference cells needs a higher bias.

(d) The notebook tests $136{,}406$ cells of an empty map. How many false alarms do you expect at each design P_{fa}, and what scatter should you allow on each count?

2. The CFAR loss. For a target that fluctuates from map to map, Swerling's first case in Lesson 27, the probability of detection has a closed form both with the noise power known and with cell-averaging CFAR over N independent reference cells:

$$P_d = P_{fa}^{1/(1+\mathrm{SNR})} \qquad \text{and} \qquad P_d = \left(1 + \frac{\alpha}{N(1+\mathrm{SNR})}\right)^{-N}.$$

(a) Find the SNR, in dB, that gives $P_d = 0.9$ at $P_{fa} = 10^{-6}$ with the noise known.

SNR = ____

(b) Find the same SNR for CA-CFAR with $N = 32$ and with $N = 144$, and the CFAR loss of each, the extra SNR it needs.

Detector	α	**SNR for** $P_d = 0.9$	**CFAR loss**
Noise known			–
CA-CFAR, $N = 32$			
CA-CFAR, $N = 144$			

3. Masking. A weak target sits four range cells nearer than a strong one, at almost the same velocity, and 8.3 dB below it. With the Hann window, the strong target's peak cell and its eight neighbors hold about 2.25 times its peak power, and all nine lie in the weak target's reference ring of 144 cells. Ignore the noise beside them.

(a) Find the CA threshold at the weak target, in dB relative to the strong target's peak, with the 10^{-6} bias. Is the weak target found?

(b) Greatest-of and smallest-of average the nearer and the farther half of the ring separately, 68 cells each. Which half holds the strong target? Predict whether each method finds the weak target, at the same bias.

4. The bias and the range. In Procedure B you walk out until the threshold loses you, then raise the bias by 3 dB and walk again. Assume a higher bias needs the same increase in SNR for the same P_d.

Predict the ratio of the two farthest ranges, and the second range if you were detected out to 5.0 m at your setting.

ratio = ____

range = ____

Part 2: Measurements (your kit, and the notebook)

Follow the procedure on the L30 page. Transmit antenna on **OUT2**, connected before you power the kit. Point the kit along a clear line and walk slowly. Every row below can also be filled from the course's recorded data: if you have no kit, or the app's overlay is not available, run the notebook's Procedure A to C cells as the page's *No Hardware?* frame describes, and mark which rows are recorded rather than your own.

Kit: _____ Overlay averages power or dB? _____ Recorded data? _____

5. **Table A: placing the threshold** (Procedure A). You at 2 m, holding still; or, recorded, the hand in chirp 0 of `L30_wall_hand.npz`.

Quantity	Your setting	Recorded data
Guard cells a side		
Reference cells a side		
Bias		
Your peak over the threshold		
Highest noise peak under it		
Threshold at your peak, 0 guard cells		

6. **Table B: the walk** (Procedure B), threshold applied; or, recorded, `L30_walker.npz`, 70 updates.

Bias	Updates that find you	Farthest range	Ratio to row 1	Updates with another crossing
Your setting			–	
Your setting $+3$ dB				

7. **Table C: false alarms on an empty scene** (Procedure C), 30 updates each; or, recorded, the first 30 chirps of `L30_empty.npz`.

Bias	Updates with a crossing	Where they land	Recorded data
Your setting			
-3 dB			
-6 dB			

Empty-scene capture: _____ Hand capture: _____

8. **Table D: notebook task (a)**, the false-alarm rate on `L30_empty.npz` or your own empty burst. 2 guard and 4 reference cells each way.

Run	`BIAS_DB`	**Design count**	**Measured**	**Measured / design**
Hann, 10^{-3}				
Hann, 10^{-4}				
Rectangular, 10^{-3}				

Your `BIAS_DB` line:

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9. **Table E: notebook task (b)**, the three methods at an 11.6 dB bias. Give the number of cells in each cluster, or "missed".

Scene	**Cluster**	`"ca"`	`"go"`	`"so"`
Wall and hand	leakage, 0 m			
	hand			
	wall			
	anywhere else			
Weak and strong	strong target			
	weak target			
	anywhere else			

10. **Table F: notebook task (c)**, the probability of detection for the hand at 12 dB SNR over 50 maps, and the detection list.

`BIAS_DB_C`	**Measured** P_d	**Noise known**	**CA-CFAR,** $N = 144$
11.6 dB			
_____ dB			

Range	**Velocity**	**SNR**	**Cells**	**Target or false alarm?**

Part 3: Writeup (individual, after the bench)

11. **The false-alarm rate you measured.** Table D's Hann rows come out above the design, and the rectangular row does not, although the noise, the window size, and the bias are the same. Explain the difference, and say what it means for a design of 10^{-6} on Hann-windowed maps.

12. **The three methods.** From Table E and your plots: why did greatest-of leave the wall's cluster unchanged, why did smallest-of grow every cluster, and why did only smallest-of find the weak target? Describe a scene in which greatest-of would change the count at a clutter edge.

13. **Detection against Lesson 27.** Compare Table F's measured P_d with the predictions, and Table B's range ratio with Question 4. Which bias would you run for the L31 Lab, and what does it cost?

14. **What you carry forward.** List what you keep from this lab for the L31 Lab and for the final report.

Documentation: _____